LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 8 Test: Heredity

Chapter / Topic: Heredity

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SECTION A

Q1. Tall pea plants with violet flowers were crossed with short pea plants with white flowers. All the progeny had violet flowers, but almost half of… **[1]**

Answer: (c) TtWW. Half the progeny are short, so the parent is Tt. All are violet, so the parent is WW.

Q2. In human beings, the sex of the child is decided by: **[1]**

Answer: (b) the father. The sperm carries either an X or a Y chromosome.

Q3. In Mendel's monohybrid cross, the phenotypic ratio in the F_2 generation is: **[1]**

Answer: (d) 3 : 1

Q4. Genes are located on the: **[1]**

Answer: (a) chromosomes (as segments of DNA)

Q5. When a pure tall (TT) pea plant is crossed with a pure short (tt) plant, the F_1 plants are: **[1]**

Answer: (c) all tall (Tt)

Q6. **Assertion (A):** Human females have a perfect pair of sex chromosomes, both called X. **Reason (R):** All children inherit an X chromosome from their… **[1]**

Answer: (b). Both statements are true, but R is a consequence of A, not its explanation.

SECTION B

Q7. What are dominant and recessive traits? Give one example of each from Mendel's experiments. **[2]**

Answer: A dominant trait is expressed even when only one copy of its gene is present (e.g. tallness, T). A recessive trait is expressed only when both copies are of the recessive type (e.g. shortness, tt).

- Dominant with example: 1
- Recessive with example: 1

OR, answer: A gene is a section of DNA that carries the information to make a protein, often an enzyme. Plant height depends on a growth hormone made with the help of an enzyme. If the gene makes an efficient enzyme, more hormone is made and the plant is tall. If the gene is altered and the enzyme is less efficient, less hormone is made and the plant is short.

- Gene → protein/enzyme: 1
- Height example: 1

Q8. How is the sex of a child determined in human beings? **[2]**

Answer: The mother has two X chromosomes, so every egg carries an X. The father has one X and one Y, so half the sperm carry X and half carry Y. If an X-carrying sperm fertilises the egg, the child is a girl (XX). If a Y-carrying sperm fertilises it, the child is a boy (XY).

- Mother XX gives X; father XY gives X or Y: 1
- XX → girl, XY → boy: 1

SECTION C

Q9. Mendel crossed pea plants with round, yellow seeds (RRYY) with plants having wrinkled, green seeds (rryy). **[3]**

(i) All round and yellow (RrYy). [1]

(ii) 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green [1]

(iii) The two traits (seed shape and seed colour) are inherited independently of each other. [1]

Q10. Show, with gametes and a checkerboard, the result of crossing two heterozygous tall pea plants (Tt × Tt). Give the phenotypic and genotypic ratios. **[3]**

Answer: Gametes from each parent: T and t. Offspring: TT, Tt, Tt, tt. Phenotypic ratio: 3 tall : 1 short. Genotypic ratio: 1 TT : 2 Tt : 1 tt.

- Gametes: 1
- Checkerboard/offspring: 1
- Both ratios: 1

OR, answer: All F_1 plants are Tt. Tallness (T) is dominant, so it masks shortness and all F_1 plants are tall. The gene for shortness is still present. When F_1 plants are self-pollinated, one in four F_2 plants gets two copies of t (tt) and is short. So the trait reappears.

- F_1 is Tt, T dominant: 1
- t is still present (not lost): 1
- F_2 tt = 1/4, short reappears: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Mendel crossed pure tall (TT) with pure short (tt) pea plants. All F_1 plants were tall (Tt), with no medium-height plants. When the F_1 plants were self-pollinated, the F_2 generation had tall and short plants in the ratio 3 : 1 (genotypes TT : Tt : tt = 1 : 2 : 1). [3]

- Parental cross: 1
- F_1 all tall: 1
- F_2 3 : 1: 1

(ii) (Any two) It has many clear contrasting visible traits (tall/short, round/wrinkled seeds, violet/white flowers). It has a short life cycle. It normally self-pollinates but can easily be cross-pollinated. It produces many seeds. [2]

- Any two: 1 + 1

OR, answer: (a) Variation means the differences among individuals of a species. It is important because some variants may survive better if the environment changes, which helps the species survive. (b) Mother (XX) × Father (XY). Gametes: egg X; sperm X or Y. Offspring: XX (girl) and XY (boy) in a 1 : 1 ratio. So the sex of the child depends on which sperm fertilises the egg.

- (a) definition 1, importance 1
- (b) gametes 1, offspring 1, conclusion 1

SECTION E

Q12. A student crossed pea plants with violet flowers (VV) with plants having white flowers (vv). All F_1 plants had violet flowers. When the F_1 plants… **[4]**

(i) Violet [1]

(ii) Vv [1]

(iii) The F_2 genotypes are VV : Vv : vv = 1 : 2 : 1, so out of 400 plants we expect 100 VV, 200 Vv and 100 vv. So about 100 of the 300 violet plants are VV. [2]

- 1 : 2 : 1 ratio: 1
- Answer 100: 1